#Maths

Continuous Probability Distributions

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Continuous Probability Distributions

1. Core Foundations: Discrete vs. Continuous & Universal Area Rules

A **random variable (X)** assigns a numerical value to each outcome of a random experiment. While a *discrete* random variable counts separated values (0, 1, 2, …), a **continuous random variable** measures any real value within an interval (such as machine lifetime, repair duration, height, or salary). For any continuous distribution, probability is always represented by the **area under the curve** over an interval—never the height at a single point.

Feature	Discrete Random Variable (Count)	Continuous Random Variable (Measure)
Possible Values	Countable values (e.g., 0, 1, 2, …)	Any real value in an interval (e.g., [a, b] or [0, ∞))
Probability Function	**PMF (Probability Mass Function): $P(X = x)$**	**PDF (Probability Density Function): $f_X(x) \geq 0$**
Exact Point Value	$P(X = a)$ can be greater than 0	$P(X = a) = 0$ (a single point has zero width/area)
Endpoint Inequality Rule	Strict (<) vs. inclusive (≤) changes the sum	$P(X < a) = P(X \leq a)$ and $P(X > a) = P(X \geq a)$
Total Probability	Sum of all PMF values: $\sum P(X = x) = 1$	Total area under PDF curve: $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$
Examples	Bernoulli, Binomial, Geometric, Poisson	**Uniform, Exponential, Normal**

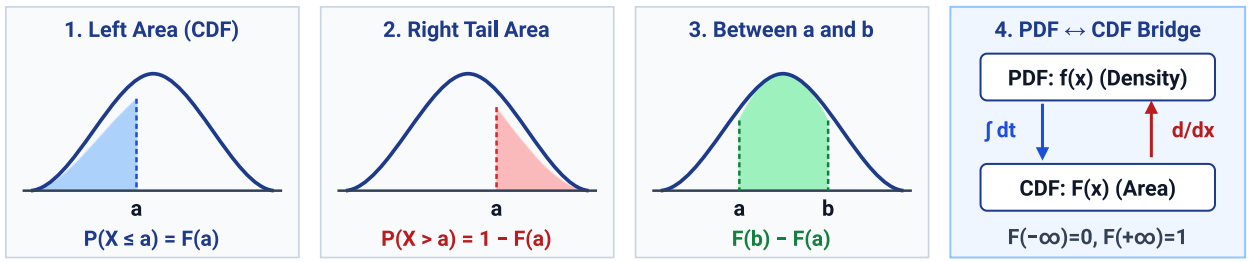


Figure 1: Universal area rules for any continuous random variable and the calculus bridge between PDF f(x) and CDF F(x).

Essential General Formulas for Any Continuous Random Variable

- **Cumulative Distribution Function (CDF):** $F_X(x) = P(X \le x) = \int_{-\infty}^{x} f_X(t)\,dt$ | **Recovering PDF from CDF:** $f_X(x) = dF_X(x) / dx$
- **Mean (Expected Value):** $\mu = E(X) = \int_{-\infty}^{\infty} x \cdot f_X(x)\,dx$
- **Variance:** $\sigma^2 = Var(X) = E(X^2) - [E(X)]^2$, where $E(X^2) = \int_{-\infty}^{\infty} x^2 \cdot f_X(x)\,dx$
- **Wording Clues in Exams:** *"At most / no more than a"* ⇒ $P(X \le a)$; *"At least / not less than a"* ⇒ $P(X \ge a)$; *"|X| < a"* ⇒ $P(-a < X < a)$ (central interval); *"|X| > a"* ⇒ $P(X < -a) + P(X > a)$ (two outer tails).

2. Continuous Uniform Distribution — X ~ U(a, b)

A continuous random variable X follows a **Uniform (Rectangular) Distribution** over $[a, b]$ when every sub-interval of equal length inside $[a, b]$ is equally likely. The PDF height remains constant at $1 / (b - a)$ across the entire support.

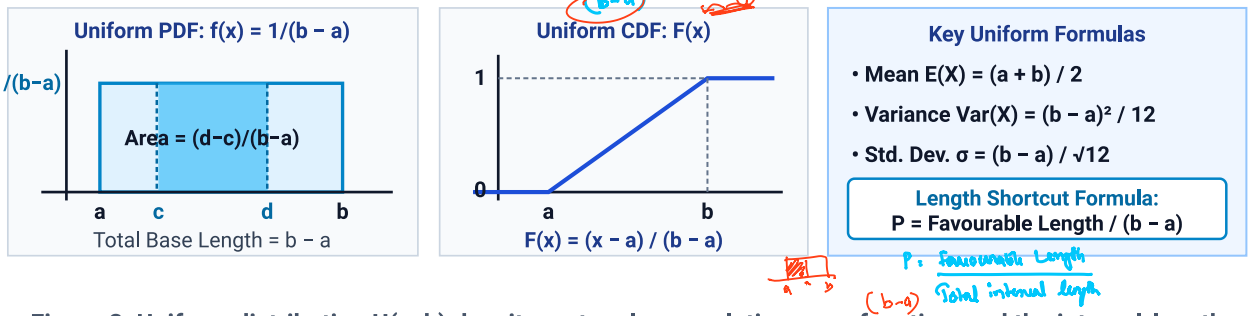
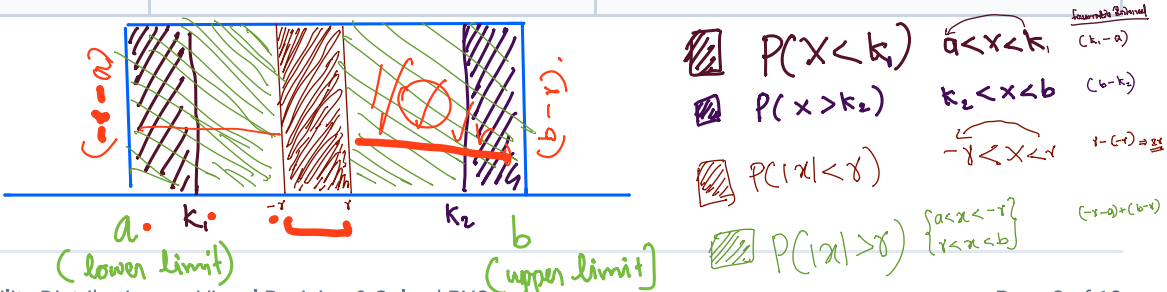


Figure 2: Uniform distribution U(a, b) density rectangle, cumulative ramp function, and the interval-length shortcut.

Question Form	Converted Range Inside [a, b]	Shortcut Formula on U(a, b)
Less than k: $P(X < k)$	$a < X < k$	$(k - a) / (b - a)$
Greater than k: $P(X > k)$	$k < X < b$	$(b - k) / (b - a)$
Absolute inner: $P(\|X\| < r)$	$-r < X < r$ (intersected with $[a, b]$)	$[r - (-r)] / (b - a) = 2r / (b - a)$
Absolute outer: $P(\|X\| > r)$	Split: $a < X < -r$ OR $r < X < b$	$[(-r - a) + (b - r)] / (b - a)$
Linear inequality: $P(gX + h > k)$	Isolate X first: $X > (k - h) / g$ (for $g > 0$)	$[b - (k - h)/g] / (b - a)$
Given Mean & Variance	Solve $(a + b)/2 = \mu$ and $(b - a)^2/12 = \sigma^2$	Example (June 2022): $E(X)=1$, $Var=3/4$ ⇒ $a+b=2$, $b-a=3$ ⇒ $U(-0.5, 2.5)$



SOLVED PYQ 1 — UNIFORM DISTRIBUTION **(M.TECH CSE 2021 · 3 MARKS)**

Question: The random variable X is uniformly distributed over $(-2, 2)$. Find: (i) $P(|X| > 1)$ (ii) $P(2X + 3 > 5)$.

Step-by-Step Solution:

- **Given Support & PDF:** $X \sim U(a, b)$ with lower limit $a = -2$ and upper limit $b = 2$. Total interval length = $b - a = 2 - (-2) = 4$. Thus, $f_X(x) = 1/4$ for $-2 < x < 2$, and 0 otherwise.
- **Part (i) Find $P(|X| > 1)$:**
  - The condition $|X| > 1$ splits into two outer regions: $X < -1$ or $X > 1$.
  - Within the support $(-2, 2)$: Left favourable interval is $(-2, -1)$ with length = $(-1) - (-2) = 1$; Right favourable interval is $(1, 2)$ with length = $2 - 1 = 1$.
  - Using the Interval Length Method: $P(|X| > 1) = P(X < -1) + P(X > 1) = (1 + 1) / 4 = 2 / 4 = 0.5$ **(or 50%)**.
- **Part (ii) Find $P(2X + 3 > 5)$:**
  - First simplify the linear inequality to isolate X: $2X + 3 > 5 \Rightarrow 2X > 5 - 3 \Rightarrow 2X > 2 \Rightarrow X > 1$.
  - Within $(-2, 2)$, the favourable interval for $X > 1$ is $(1, 2)$, which has length = $2 - 1 = 1$.
  - Therefore, $P(2X + 3 > 5) = P(X > 1) = (2 - 1) / 4 = 1 / 4 = 0.25$ **(or 25%)**.

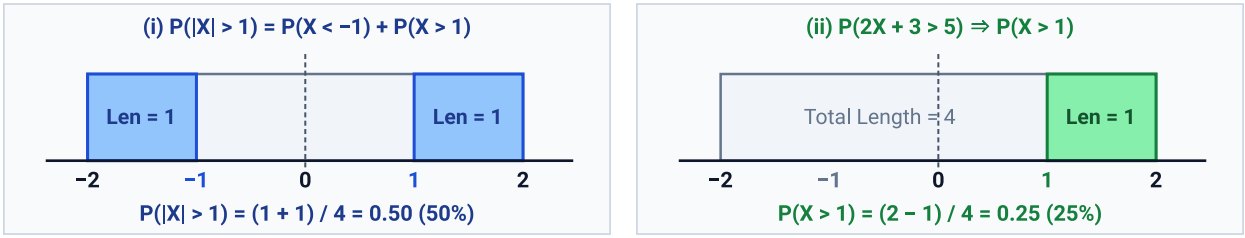


Figure 3: Shaded interval diagrams for PYQ 1 on X ~ U(−2, 2) with constant height f(x) = 1/4 and total base length 4.

3. Exponential Distribution — X ~ Exp(λ)

The **Exponential Distribution** models continuous waiting time, machine lifetime, call duration, or repair time until an event occurs at a constant rate $\lambda > 0$. It is the only continuous distribution with the **Memoryless Property**: if a component has already survived s hours, the probability it survives an additional t hours is identical to a brand-new component surviving t hours.

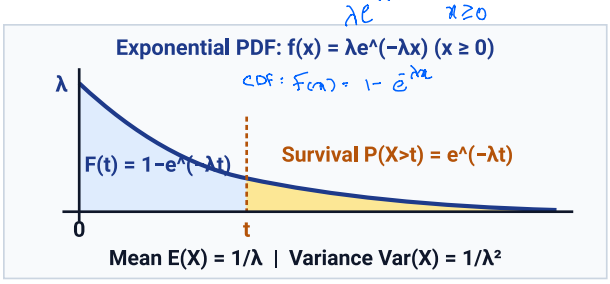


Figure 4: Exponential distribution PDF curve, survival tail formula, and memoryless property timeline.

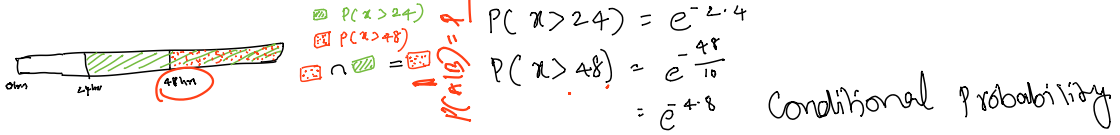
Exponential Question Wording	Probability Notation	Formula (Given Rate λ = 1 / Mean)
Exceeds / survives beyond / at least t	$P(X > t)$ or $P(X \geq t)$	$e^{-\lambda t}$ (Survival Function)
Less than / fails before / at most t	$P(X < t)$ or $P(X \leq t)$	$F(t) = 1 - e^{-\lambda t}$ (CDF)
Between two times a and b	$P(a < X < b)$	$e^{-\lambda a} - e^{-\lambda b}$
Survived s; works for another t	$P(X > s + t \mid X > s)$	$P(X > t) = e^{-\lambda t}$ (Memoryless Property)

SOLVED PYQ 2 — EXPONENTIAL DISTRIBUTION (M.TECH CSE · 221TCS002 · JUNE 2026 · 7 MARKS)

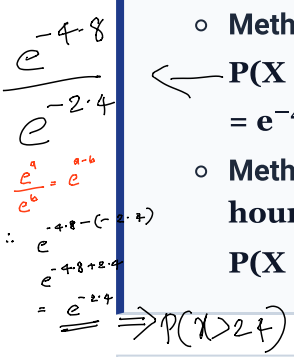
Question: A new machine is put into operation at time zero. Its lifetime is an exponential random variable with mean life 10 hours. **(i)** What is the probability that the machine will work continuously for one day? **(ii)** If it has not failed by the end of the first day, what is the probability that it will work for the whole of the next day?

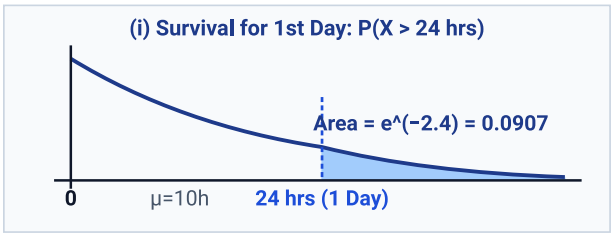
Step-by-Step Solution:

- **Define Variable & Rate Parameter (λ):** Let X be the lifetime of the machine in hours, so $X \sim Exp(\lambda)$. Given mean life $E(X) = 1/\lambda = 10$ hours $\Rightarrow \lambda = 1/10 = 0.1$ per hour. The PDF is $f_X(x) = 0.1\, e^{-0.1x}$ for $x \geq 0$.
- **Part (i) Probability of working continuously for one day (1 day = 24 hours):**
  - Required probability: $P(X > 24)$. Using the exponential survival formula $P(X > t) = e^{-\lambda t}$.
  - $P(X > 24) = \int_{24}^{\infty} 0.1\, e^{-0.1x}\, dx = e^{-0.1 \times 24} = e^{-2.4} \approx 0.0907$ (or 9.07%).



- **Part (ii) Probability of working for the whole next day ($X > 48$), given it survived the first day ($X > 24$):**
  - **Method 1 (Conditional Probability Formula):** Since $(X > 48) \cap (X > 24) = (X > 48)$:
    $P(X > 48 \mid X > 24) = P(X > 48) / P(X > 24) = e^{-0.1 \times 48} / e^{-0.1 \times 24} = e^{-4.8} / e^{-2.4} = e^{-4.8 - (-2.4)} = e^{-2.4} \approx 0.0907$ **(or 9.07%)**.
  - **Method 2 (Direct Memoryless Property):** With $s = 24$ **hours** and additional time $t = 24$ **hours**:
    $P(X > 24 + 24 \mid X > 24) = P(X > 24) = e^{-0.1 \times 24} = e^{-2.4} \approx 0.0907$ **(or 9.07%)**.





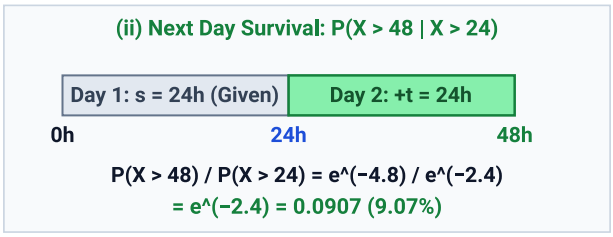


Figure 5: Visual representation of PYQ 2 showing survival beyond 24 hours and memoryless survival from 24 to 48 hours.

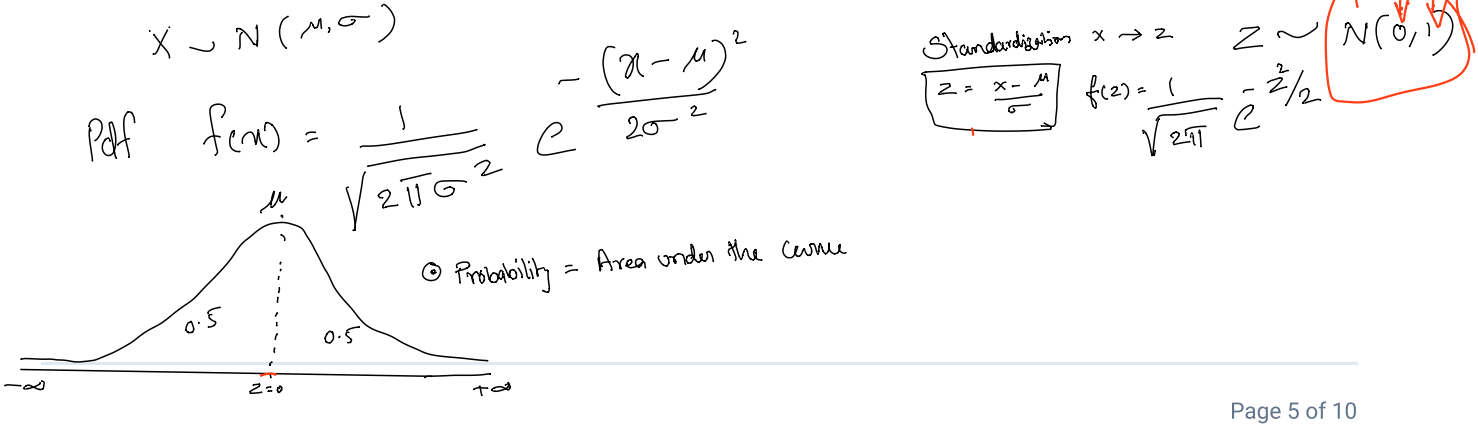
4. Normal Distribution — $X \sim N(\mu, \sigma^2)$ & Two Types of Exam Questions

The **Normal (Gaussian) Distribution** is a continuous, symmetric, bell-shaped distribution centered at mean μ with spread controlled by standard deviation σ. Its PDF is $f(x) = [1 / (\sigma\sqrt{(2\pi)})] \cdot e^{-(x-\mu)^2 / (2\sigma^2)}$ for $-\infty < x < \infty$. To solve any problem, we standardize X into a **Standard Normal Variable** $Z \sim N(0, 1)$ and use the KTU Standard Normal Z-Table, which gives the area between 0 and z: $A = P(0 < Z < z)$.

> $Z = (X - \mu) / \sigma \iff X = \mu + Z\sigma$ | **Total Area = 1** (Left half = 0.50, Right half = 0.50) | **CDF Rule:** $P(X < a) = F(a)$, $P(X > a) = 1 - F(a)$

Approach 1: Z → Probability (Direct Graph Method — Given Z, Find Probability)

When to use: Use when X or Z is given and you need to find the probability. **Steps: (1)** Standardize X to Z, draw the bell curve, and shade the required region. **(2)** Find the table area $A = P(0 < Z < |z|)$ from the Z-table. **(3)** Compute the final probability using the 4 visual graph rules below:



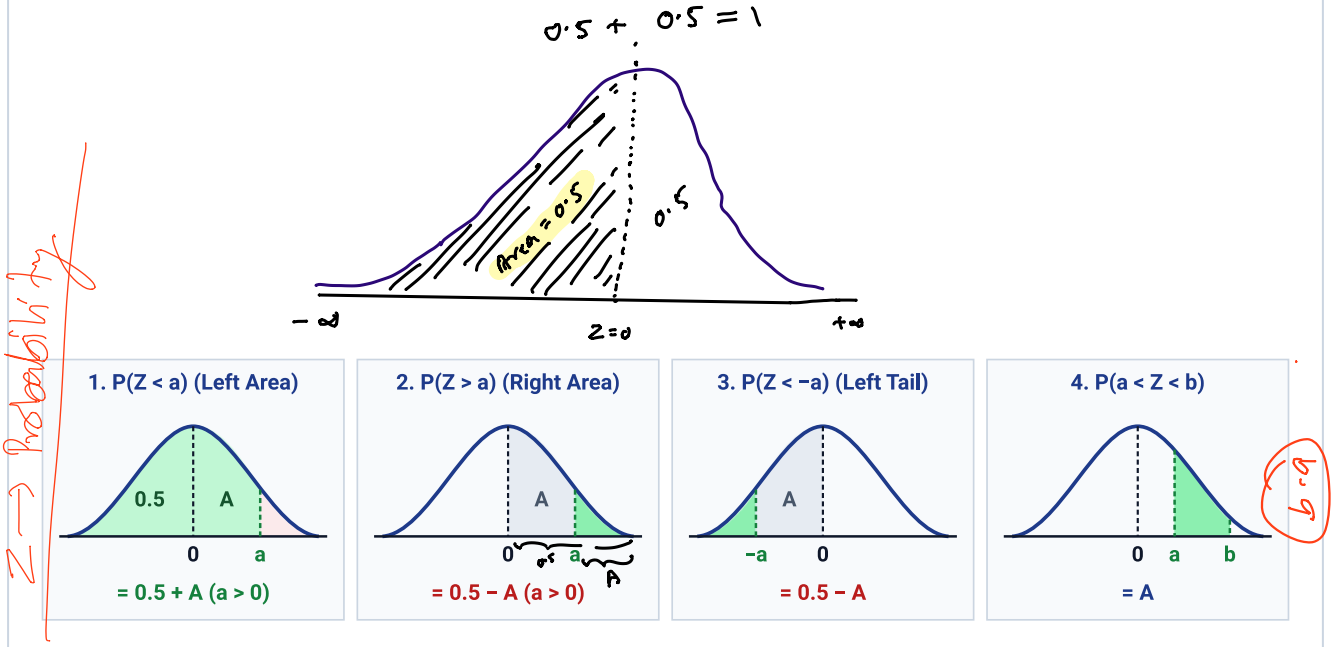


Figure 6A: Approach 1 (Z → Probability) Direct Graph Rules using 0.50 and Z-table area A = P(0 < Z < |z|).

- **Worked Examples for Approach 1 (Direct Graph Method):**
  - **Example 1 — Find $P(Z > 0.85)$:** From Z-table at $z = 0.85$, $A = 0.3023$. Since we need the right tail ($a > 0$), $P(Z > 0.85) = 0.50 - A = 0.50 - 0.3023 = 0.1977$.
  - **Example 2 — Find $P(-2.08 < Z < 2.08)$:** From Z-table at $z = 2.08$, $A = 0.4812$. By symmetry across 0, $P(-2.08 < Z < 2.08) = 2A = 2 \times 0.4812 = 0.9624$.

Approach 2: Probability → Z (CDF / Inverse Z-Table Method — Given Probability, Find Z)

When to use: Use when a probability (or percentage) is given and you must find the cutoff Z (or X). **4-Step Rule:**

1. **Convert to Left Cumulative Probability $P(Z \le z)$:** If given a right-tail probability $P(Z \ge z) = P$, convert to left area using $P(Z \le z) = 1 - P$.
2. **Find Table Area A:** Take the absolute difference from 0.50: $A = |P(Z \le z) - 0.50|$.
3. **Look Up A in the Z-Table to Get $|z|$:** Locate the value closest to A inside the Z-table body and read off $|z|$.
4. **Assign the Sign of Z:** Attach + **(positive)** if $P(Z \le z) > 0.50$; attach − **(negative)** if $P(Z \le z) < 0.50$. Then compute $X = \mu + Z\sigma$.

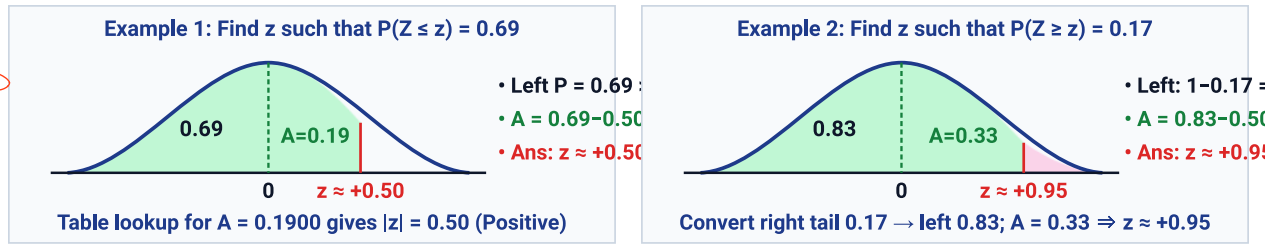


Figure 6B: Approach 2 (Probability → Z) worked examples from lecture notes showing left cumulative conversion and table area A.

Question Type	Method / Approach	Key Rule & Formula Summary (Where Z-Table Gives A = P(0 < Z < |z|))
Type 1: Z (or X) is given, find Probability	**Direct Graph Method** (Z → Probability)	Use 0.50 and table area A: • $P(Z < a) = 0.50 + A$ (for $a > 0$) or $0.50 - A$ (for $a < 0$) • $P(Z > a) = 0.50 - A$ (for $a > 0$) or $0.50 + A$ (for $a < 0$) • $P(a < Z < b) = A_b - A_a$ (same side of 0) or $A_a + A_b$ (opposite sides of 0)
Type 2: Probability is given, find Z (or X)	**CDF / Inverse Z-Table Method** (Probability → Z)	**1.** Convert to left cumulative $P = P(Z \le z)$ (if right tail p is given, use $P = 1 - p$). **2.** Compute table area $A = \|P - 0.50\|$ and look up $\|z\|$ in the Z-table. **3.** Assign sign: $+z$ if $P > 0.50$; $-z$ if $P < 0.50$. Then $X = \mu + Z\sigma$.

SOLVED PYQ 3A — NORMAL DISTRIBUTION QUANTILES (M.TECH CSE · JUNE 2026 · 7 MARKS)

Question: X is a normal random variable with mean 50 and standard deviation 10. Find α and β such that $P(X < \alpha) = 0.1$ and $P(X > \beta) = 0.05$.

Step-by-Step Solution:

- **Given Parameters:** $X \sim N(\mu = 50, \sigma^2 = 10^2)$, so $\mu = 50$, $\sigma = 10$, and $Z = (X - 50) / 10$.
- **Part 1 — Find α such that $P(X < \alpha) = 0.10$:**
  - **Step 1 (Check Left Area):** $P(X < \alpha) = 0.10$ is already a left cumulative area. Since $0.10 < 0.50$, α lies to the left of the mean 50, so Z_1 is **negative**.
  - **Step 2 (Table Area A_1):** $A_1 = 0.50 - 0.10 = 0.4000$.
  - **Step 3 (Z-Table Lookup):** Looking up $A_1 = 0.4000$ in the Standard Normal Table (where $Z = 1.28$ gives 0.3997), we get $|Z_1| = 1.28 \Rightarrow Z_1 = -1.28$.
  - **Step 4 (Solve for α):** $(\alpha - 50) / 10 = -1.28 \Rightarrow \alpha = 50 + 10(-1.28) = 50 - 12.8 = 37.2$.

- **Part 2 — Find β such that $P(X > \beta) = 0.05$:**
  - **Step 1 (Convert Right Tail to Left Cumulative):** $P(X > \beta) = 0.05 \Rightarrow P(X \leq \beta) = 1 - 0.05 = 0.95$. Since $0.95 > 0.50$, β lies to the right of the mean 50, so Z_2 is **positive**.
  - **Step 2 (Table Area A_2):** $A_2 = 0.95 - 0.50 = 0.4500$.
  - **Step 3 (Z-Table Lookup):** In the Z-table, $Z = 1.64$ gives 0.4495 and $Z = 1.65$ gives 0.4505, so $Z_2 = +1.645$ (or $+1.64$).
  - **Step 4 (Solve for β):** $(\beta - 50) / 10 = 1.645 \Rightarrow \beta = 50 + 10(1.645) = 50 + 16.45 = 66.45$ (or 66.4 using 1.64).

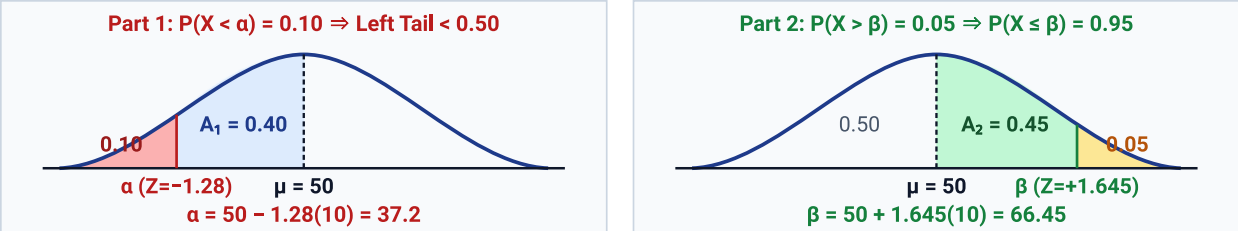


Figure 7: Shaded bell curve solutions for PYQ 3A showing table areas $A_1 = 0.40$ ($Z_1 = -1.28$) and $A_2 = 0.45$ ($Z_2 = +1.645$).

SOLVED PYQ 3B — NORMAL DISTRIBUTION INVERSE PROBLEM (M.TECH CSE · NOV/DEC 2025 · 7 MARKS)

Question: The monthly salaries of employees in a large company are normally distributed. Employees earning Rs. 60,000 or less are in Category I; those earning between Rs. 60,000 and Rs. 80,000 are in Category II; the rest are in Category III. If 69% are in Category I and 14% in Category II, what is the average salary in the company?

Step-by-Step Solution:

- **Define Variable & Cumulative Left Areas:** Let monthly salary $X \sim N(\mu, \sigma^2)$, where μ is the required average salary.
  - Category I ($X \leq 60{,}000$): Left cumulative probability $P(X \leq 60{,}000) = 69\% = 0.69$.
  - Category II ($60{,}000 < X \leq 80{,}000$): $P(60{,}000 < X \leq 80{,}000) = 14\% = 0.14$.
  - Combining Category I + Category II gives the cumulative probability up to Rs. 80,000: $P(X \leq 80{,}000) = 0.69 + 0.14 = 0.83$ (and Category III is the right tail $P(X > 80{,}000) = 1 - 0.83 = 0.17$).
- **Find Z_1 for Cut-off $x_1 = 60{,}000$:**
  - Since $P(X \leq 60{,}000) = 0.69 > 0.50$, Z_1 is **positive**, and table area $A_1 = 0.69 - 0.50 = 0.1900$.
  - From the Z-table ⟶ $(A = 0.1915)$, $Z_1 = +0.50$.
  - Standardizing: $(60{,}000 - \mu) / \sigma = 0.50 \Rightarrow 60{,}000 - \mu = 0.50\sigma$ **--- [Equation 1]**

- **Find Z_2 for Cut-off $x_2 = 80{,}000$:**
  - Since $P(X \le 80{,}000) = 0.83 > 0.50$, Z_2 is **positive**, and table area $A_2 = 0.83 - 0.50 = 0.3300$.
  - From the Z-table (Page 23 of notes, $A = 0.3289$), $Z_2 = +0.95$.
  - Standardizing: $(80{,}000 - \mu) / \sigma = 0.95 \Rightarrow 80{,}000 - \mu = 0.95\sigma$ **--- [Equation 2]**
- **Solve Simultaneous Equations [1] & [2] for σ and Average Salary μ:**
  - Subtract Equation [1] from Equation [2]: $(80{,}000 - \mu) - (60{,}000 - \mu) = 0.95\sigma - 0.50\sigma \Rightarrow 20{,}000 = 0.45\sigma \Rightarrow \sigma = 20{,}000 / 0.45 = \text{Rs. } 44{,}444.44$.
  - Substitute σ into Equation [1]: $\mu = 60{,}000 - 0.50(44{,}444.44) = 60{,}000 - 22{,}222.22 = \text{Rs. } 37{,}777.78$.

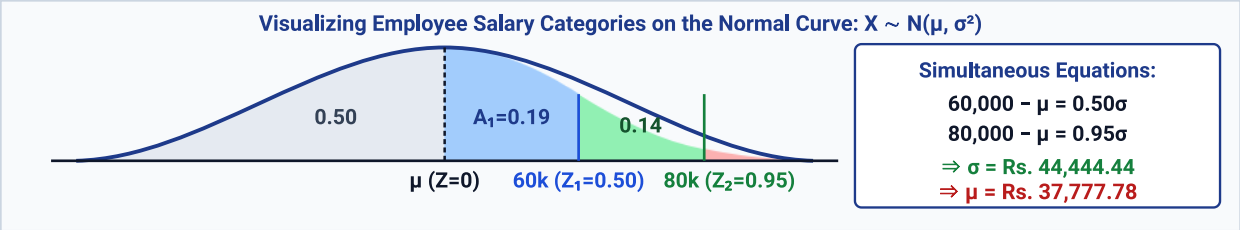
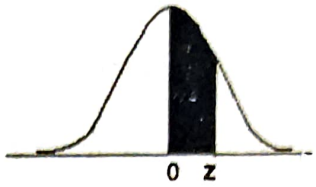


Figure 8: Visual decomposition of PYQ 3B showing $A_1 = 0.69 - 0.50 = 0.19$ ($Z_1 = 0.50$) and $A_2 = 0.83 - 0.50 = 0.33$ ($Z_2 = 0.95$).

5. Master Formula & Exam Identification Cheat Sheet

Distribution	Exam Clues / Use	PDF f(x) & Support	Mean & Variance	Core Probability Formulas
Uniform $X \sim U(a, b)$	"Uniformly distributed", equal chance in $[a, b]$	$f(x) = 1 / (b - a)$ for $a \le x \le b$	$E(X) = (a + b) / 2$ $Var(X) = (b - a)^2 / 12$	$P(c < X < d) = (d - c) / (b - a)$ $F(x) = (x - a) / (b - a)$
Exponential $X \sim Exp(\lambda)$	Waiting time, machine lifetime, repair time	$f(x) = \lambda e^{-\lambda x}$ for $x \ge 0$ ($\lambda > 0$)	$E(X) = 1 / \lambda$ $Var(X) = 1 / \lambda^2$	$P(X > t) = e^{-\lambda t}$ | $P(X \le t) = 1 - e^{-\lambda t}$ $P(a < X < b) = e^{-\lambda a} - e^{-\lambda b}$ Memoryless: $P(X > s+t \mid X > s) = e^{-\lambda t}$
Normal $X \sim N(\mu, \sigma^2)$	Bell curve, heights, salaries, scores, quantiles	$[1/(\sigma\sqrt{(2\pi)})]e^{-(x-\mu)^2/(2\sigma^2)}$ for $-\infty < x < \infty$	$E(X) = \mu$ $Var(X) = \sigma^2$	Standardize: $Z = (X - \mu) / \sigma$ Use $0.50 \pm A$ from Z-table; Inverse: $A = \lvert P(X \le x) - 0.50 \rvert$

AREA UNDER STANDARD NORMAL CURVE



NORMAL DISTRIBUTION

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.01595	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.07535
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.22575	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2518	0.2549
0.7	0.2580	0.26115	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.29955	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.44845	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.48645	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.49795	0.4980	0.4981
2.9	0.4981	0.4982	0.49825	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.49865	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.49975	0.4998
3.5	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998
3.6	0.4998	0.4998	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.7	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.8	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.49995	0.49995	0.49995
3.9	0.49995	0.49995	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000